LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Annual Exam Practice Paper 1 (CBSE pattern)

Chapter / Topic: Motion · Force and Laws of Motion · Work and Energy · Sound · Matter in Our Surroundings · Is Matter Around Us Pure? · Structure of the Atom · The Cell · Tissues

Name: _____ Roll No.: _____ Date: _____

Time: 3 hours **Max. Marks: 80**

General instructions:

1. This paper has 39 questions in 5 sections (A–E). All questions are compulsory. Internal choice is given in some questions.
2. Section A: Q1–Q20, 1 mark each (MCQs and Assertion–Reason). Section B: Q21–Q26, 2 marks each. Section C: Q27–Q33, 3 marks each. Section D: Q34–Q36, 5 marks each. Section E: Q37–Q39, case-based, 4 marks each.
3. Show formulae, substitution and units in every numerical. Draw neat, labelled diagrams wherever asked.
4. This paper is topic-based. Check the topic list against your school's annual exam syllabus before using it.

SECTION A

20 marks

Q1–Q16: choose the correct option. Q17–Q20: Assertion–Reason. For these, choose (a) both A and R true and R explains A; (b) both true but R does not explain A; (c) A true, R false; (d) A false, R true.

Q1. The slope of a distance–time graph gives the: **[1]**

(a) acceleration (b) displacement
(c) force (d) speed

Q2. The velocity of a car changes from 10 m/s to 20 m/s in 5 s. Its acceleration is: **[1]**

(a) 0.5 m/s^2 (b) 2 m/s^2
(c) 4 m/s^2 (d) 6 m/s^2

Q3. The momentum of a 2 kg ball moving at 3 m/s is: **[1]**

(a) 1.5 kg m/s (b) 5 kg m/s
(c) 6 kg m/s (d) 9 kg m/s

Q4. Which of these is the SI unit of work? **[1]**

(a) joule (b) newton
(c) watt (d) pascal

Q5. The range of frequencies that a normal human ear can hear is about: **[1]**

(a) 2 Hz to 2,000 Hz (b) 20 Hz to 20,000 Hz
(c) 200 Hz to 2,00,000 Hz (d) 20 kHz to 200 kHz

Q6. A wave has a time period of 0.02 s. Its frequency is: **[1]**

(a) 0.02 Hz (b) 20 Hz
(c) 50 Hz (d) 200 Hz

Q7. Which of these changes directly from solid to vapour on heating? **[1]**

(a) camphor (b) sugar
(c) common salt (d) sand

Q8. In which of these are the forces of attraction between particles the strongest? **[1]**

(a) oxygen gas (b) water
(c) kerosene (d) an iron nail

Q9. Which of these is a colloid? **[1]**

(a) salt solution (b) sugar solution
(c) chalk powder in water (d) milk

Q10. The electronic configuration of an atom is 2, 8, 3. Its valency is: **[1]**

(a) 1 (b) 2
(c) 3 (d) 5

Q11. Isobars have the same: **[1]**

(a) mass number (b) atomic number
(c) number of protons (d) number of electrons

Q12. Which cell organelle packages materials and dispatches them to places inside and outside the cell? **[1]**

(a) ribosome (b) Golgi apparatus
(c) lysosome (d) nucleus

Q13. Which plant tissue gives flexibility to leaf stalks and young stems? **[1]**

(a) collenchyma (b) parenchyma
(c) sclerenchyma (d) xylem

Q14. A bacterial cell does NOT have: **[1]**

(a) a cell membrane
(b) a nucleus with a nuclear membrane
(c) cytoplasm
(d) ribosomes

Q15. Tendons connect: **[1]**

(a) bone to bone (b) muscle to muscle
(c) muscle to bone (d) skin to muscle

Q16. An animal cell is placed in a strong salt solution. It will: **[1]**

(a) swell up (b) burst
(c) stay the same (d) shrink

Q17. **Assertion (A):** A passenger standing in a bus tends to fall backwards when the bus starts suddenly.
Reason (R): The feet move forward with the bus, but the upper body tends to stay at rest because of inertia. **[1]**

Q18. **Assertion (A):** Sound travels faster in steel than in air.
Reason (R): Sound travels fastest in a vacuum. **[1]**

Q19. **Assertion (A):** The isotopes of an element have similar chemical properties.
Reason (R): Chemical properties depend on the number of valence electrons, which is the same for all isotopes of an element. **[1]**

Q20. **Assertion (A):** Sclerenchyma cells are living cells.
Reason (R): The walls of sclerenchyma cells are thickened with lignin. **[1]**

SECTION B

12 marks

Q21. A bus starts from rest and reaches a speed of 54 km/h in 10 s. Assuming uniform acceleration, find (a) its acceleration and (b) the distance it covers in this time. **[2]**

Q22. The velocity of a 5 kg object increases from 3 m/s to 7 m/s in 2 s. Find the change in its momentum and the force acting on it. **[2]**

OR

A box of mass 10 kg is lifted to a shelf 2 m high. How much work is done against gravity? ($g = 9.8\ m/s^2$) Where does this energy go?

Q23. Convert: (a) 300 K into °C, (b) 573 K into °C, (c) 25 °C into K. (d) What is the physical state of water at 573 K? **[2]**

Q24. How much sugar and how much water are needed to make 200 g of a 15% (mass by mass) sugar solution? **[2]**

Q25. Where in the cell are (a) chromosomes and (b) ribosomes found? Give the function of each. **[2]**

Q26. Name the tissue: (a) that stores fat below the skin, (b) that joins two bones, (c) that lines the kidney tubules, (d) that forms the husk of a coconut. **[2]**

SECTION C

21 marks

Q27. A car moves at a steady 10 m/s for 4 s. The brakes are then applied and it slows down uniformly, stopping 6 s later.

(a) Describe the shape of its velocity–time graph.

(b) Find the total distance covered.

(c) Find the retardation. **[3]**

Q28. An object of mass 8 kg moves at 5 m/s.

(a) Find its kinetic energy.

(b) What will its kinetic energy be if its speed is doubled?

(c) How much work must be done to stop it when it is moving at 5 m/s? **[3]**

Q29. A man stands between two parallel cliffs and claps once. He hears the first echo after 2 s and the second echo after 3 s. Find the distance between the two cliffs. (Speed of sound = 340 m/s) **[3]**

OR

What are loudness and pitch? On which property of a sound wave does each depend? Give one example of a high-pitched and one of a low-pitched sound.

Q30. How do (a) surface area, (b) humidity and (c) wind speed affect the rate of evaporation? Give an everyday example for each. **[3]**

Q31. A sodium atom has atomic number 11 and mass number 23.

(a) How many protons, neutrons and electrons does it have?

(b) Write its electronic configuration and draw its Bohr model.

(c) What is its valency? Why? **[3]**

Q32. Name the method you would use to separate: (a) iron filings from sand, (b) butter from curd, (c) tea leaves from tea. Give a reason for each choice. **[3]**

Q33. Compare striated (skeletal), smooth and cardiac muscle under these headings: (a) where they are found, (b) voluntary or involuntary, (c) shape of the cells. **[3]**

SECTION D

15 marks

Q34. (a) Write the three equations of motion for uniformly accelerated motion.

(b) A ball is thrown straight up with a velocity of 20 m/s. Find the maximum height it reaches and the time it takes to get there. (Take $g = 10\ m/s^2$.)

(c) With what speed will it come back to the thrower's hand? **[5]**

OR

(a) State the law of conservation of momentum.

(b) A bullet of mass 25 g is fired from a gun of mass 5 kg with a velocity of 400 m/s. Find the recoil velocity of the gun.

(c) Why is the gun's recoil speed so much smaller than the bullet's speed?

Q35. (a) Describe Rutherford's alpha-particle scattering experiment. Give three observations and the conclusion drawn from each.
(b) What was the main drawback of Rutherford's model?
(c) How did Bohr's model overcome this drawback? **[5]**

OR

(a) What are isotopes? Give two uses of isotopes.
(b) Atom X has 6 protons and 6 neutrons; atom Y has 6 protons and 8 neutrons. Find their atomic and mass numbers. How are X and Y related?
(c) What are isobars? Give one pair as an example.

Q36. (a) Draw a neat, labelled diagram of a plant cell.
(b) Give three differences between a plant cell and an animal cell. **[5]**

OR

(a) What are complex permanent tissues? Name the elements that make up xylem and phloem.
(b) Give the function of xylem and of phloem.
(c) Which element of phloem is made of dead cells?

SECTION E

12 marks

Read each passage and answer the questions that follow.

Q37. Cars have seat belts and airbags to protect passengers in a crash. When a car stops suddenly, the passengers keep moving forward. A seat belt holds them back, and an airbag inflates in a fraction of a second to cushion them. In one crash test, a car moving at 20 m/s stopped in 0.1 s, and a 60 kg dummy inside was brought to rest with it. **[4]**

(i) Why do passengers keep moving forward when the car stops suddenly? **[1]**

(ii) Find the force needed to stop the 60 kg dummy in 0.1 s. **[2]**

(iii) How does an airbag reduce the force on a passenger? **[1]**

Q38. Seawater contains about 35 g of salts in every 1 kg of seawater. In salt pans along the Gujarat coast, seawater is let into shallow beds and left in the sun. The water evaporates and salt is left behind. To get pure crystals of a substance such as copper sulphate in the laboratory, crystallisation is used instead of heating the solution to dryness. **[4]**

(i) Find the concentration of salts in seawater in mass by mass percent. **[2]**

(ii) Why are the salt beds kept shallow? **[1]**

(iii) Why is crystallisation better than heating a solution to dryness for getting a pure solid? **[1]**

Q39. Meera put a freshly cut white flower with a long stalk into a glass of water mixed with red ink. After a few hours, red streaks appeared in the petals. When she cut the stalk across and looked at it with a hand lens, she saw red dots arranged in a ring. **[4]**

(i) Which tissue carried the coloured water up the stalk? **[1]**

(ii) Name the two kinds of cells in this tissue that actually carry the water. **[1]**

(iii) Which tissue carries food in a plant? How is its direction of transport different from that of the tissue in (i)? **[2]**

~ All the best for your annual exam! ~